

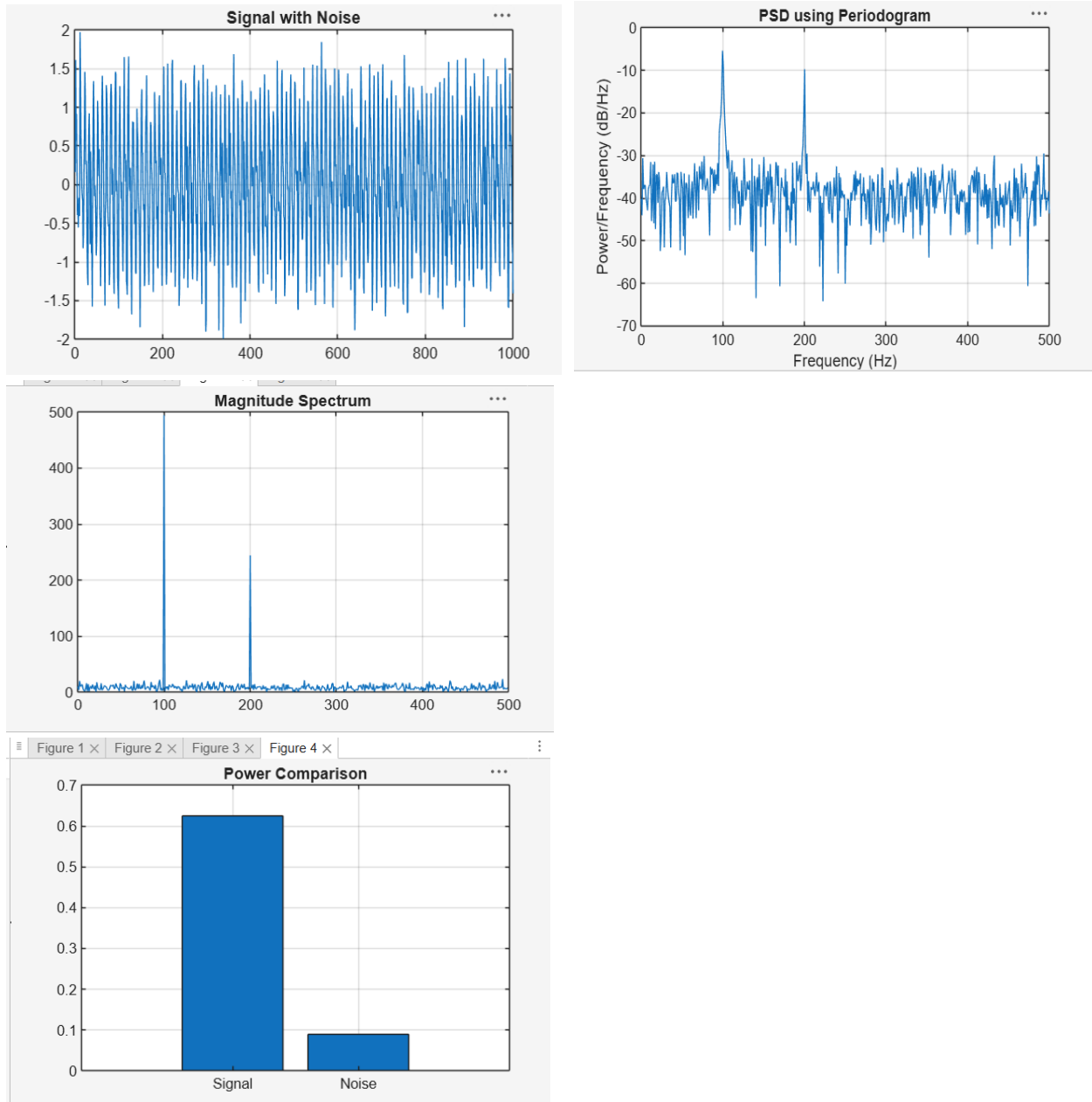
Experiment 7

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Periodogram Method in MATLAB

Code:

```
clc; clear; close all fs=1000; t=(0:999)/fs; s=sin(2*pi*100t)+0.5sin(2*pi*200t);  
n=0.3*randn(size(t)); x=s+n; figure, plot(x), grid on title('Signal with Noise') figure,  
periodogram(x,[],[],fs) title('PSD using Periodogram') figure X=abs(fft(x));  
f=(0:length(X)-1)*fs/length(X); plot(f(1:500),X(1:500)), grid on title('Magnitude  
Spectrum') figure bar([mean(s.^2) var(n)])  
set(gca,'XTickLabel',{'Signal','Noise'}) title('Power Comparison') grid on
```

Output:



7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, Hanning Windows in MATLAB

Code:

```
clc; clear; close all; % Signal fs = 1000; N = 256; t = (0:N-1)/fs; x = sin(2*pi*100*t) +
0.8*sin(2*pi*110*t); NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT); % Windows w = {rectwin(N),
hamming(N), hann(N)}; names = {'Rectangular','Hamming','Hanning'}; % Processing for i =
1:3 xw = x(:).*w{i}; X = fft(xw,NFFT); P{i} = abs(X(1:NFFT/2)); P{i} = P{i}/max(P{i}); end %%
Individual Spectrums figure; for i = 1:3 subplot(3,1,i); plot(f,20*log10(P{i}),'LineWidth',1.5);
```

```

title(['Spectrum using ' names{i} ' Window']); xlim([0 250]); grid on; end %% Comparison
figure;      plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on; %% Window Shapes
figure; for i = 1:3 subplot(3,1,i); plot(w{i},'LineWidth',1.5); title([names{i} ' Window']); grid on;
end %% Peak Detection for i = 1:3 [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
fprintf('\n%s Window Frequencies:\n',names{i}); disp(loc{i}); end %% Peak Marking
(Hamming)      figure;      plot(f,20*log10(P{2}),'LineWidth',1.5);      hold      on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');      title('Detected      Frequencies
(Hamming)'); xlim([50 170]); grid on; %% Main Peaks fprintf('\nMain Spectral Peaks:\n'); for i
= 1:3 [~,idx] = max(P{i}); fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx)); end

```

Output:

